Warm-up / More late parcels, a lower rate?

Compare counts with the right totals

Name: _____ Date: _____

Suggested lesson time: 30 minutes. Work individually, then compare when invited.

Useful words

Count: The number of cases, such as missed appointments.

Rate: The count divided by the relevant total; multiply by 100 to express it as a percentage.

Denominator: The number below the fraction bar: the total used as the comparison base.

Read the scenario

Invented scenario: A depot delivered 40 parcels on Monday, of which 4 were late. On Tuesday it delivered 160 parcels, of which 8 were late. A parcel is late if it arrives after its promised deadline.

Produce: Write two percentage calculations and one comparison sentence.

Your steps

1. Calculate each day's late rate: late parcels / all delivered parcels x 100.

2. State whether the late count and late rate moved in the same direction.

Before you finish: Use each day's own delivery total, not the same denominator for both days.

Your response

Main activity / The missed-appointment claim

Name: _____

Read the scenario

Invented scenario: A clinic had 100 missed appointments out of 1,000 booked visits in March and 80 out of 800 in April. It advertises: 'Missed appointments fell by 20%, so patients became more reliable.'

Produce: Show the percentage change in the count and both monthly rates, then write two sentences judging the advertisement.

Your steps

1. Calculate the percentage change in the missed count: (new count - old count) / old count x 100. Use (80 - 100) / 100 x 100.
2. Calculate each month's missed rate: missed / booked x 100.
3. Separate the count change from the rate change and judge whether the reliability claim follows from these figures.

Before you finish: Name the denominator for each month. A lower count need not mean a lower rate.

Your response

Exit ticket / More missing assignments?

Name: _____

Read the scenario

Invented scenario: A course recorded 18 missing assignments out of 120 assignments due in September, and 20 out of 200 due in October. The teacher says, 'More assignments were missing, so the missing-assignment rate got worse.'

Produce: Write both rates and two sentences assessing the teacher's conclusion.

Your steps

1. Calculate each month's missing rate: missing assignments / assignments due x 100.
2. State which month has the higher missing count and which has the higher missing rate.
3. Explain whether the claim about a worsening rate follows, without claiming that these figures identify a cause.

Before you finish: Use assignments due as the denominator; do not divide by a number of students that was never supplied.

Your response